

SMART CAMPUS EVENT MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled” Smart Campus Event Management System” by K.LOHITH VENKATA SAI REDDY(VTU26667) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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Abstract

The Smart Campus Event Management System is a web-based application developed to streamline the organization and participation of campus events such as workshops, seminars, and conferences. The system provides an integrated platform where students can easily browse events, register online, and download tickets, while administrators can manage event details and monitor participation effectively. The application is built using Spring Boot for backend development, Thymeleaf for dynamic frontend rendering, and MySQL for data storage. It incorporates secure user authentication, session management, and form validation to ensure data integrity and user privacy. A premium user interface with vintage gradient design enhances user experience, making the platform visually appealing and easy to navigate. Additionally, the system includes a simulated payment success interface, automated PDF ticket generation, and an admin dashboard with real-time data visualization using charts. These features help improve operational efficiency and provide valuable insights into event participation trends. Overall, the proposed system reduces manual workload, improves accessibility, and demonstrates the practical implementation of modern full-stack web development technologies in an academic environment.

Keywords: Smart Campus Event Management System, Spring Boot, Thymeleaf, MySQL, Web Application, User Authentication, PDF Ticket Generation, Admin Dashboard, Data Visualization

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CSS	Cascading Style Sheets
DBMS	Database Management System
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JPA	Java Persistence API
JVM	Java Virtual Machine
MVC	Model View Controller
MySQL	My Structured Query Language
OTP	One Time Password
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
UX	User Experience

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Chapter 1

INTRODUCTION

1.1 Introduction

The Smart Campus Event Management System is a web-based application designed to simplify the process of managing academic and campus events such as workshops, seminars, and conferences. In many institutions, event handling is still performed manually, which leads to inefficiency, data inconsistency, and increased workload. This system provides a centralized digital platform where students can browse available events, register online, and download tickets. At the same time, administrators can manage event details, track registrations, and analyze participation data effectively.

The application is developed using modern technologies such as Spring Boot, Thymeleaf, and MySQL to ensure scalability, security, and performance. The system also includes features like user authentication, premium user interface design, and automated PDF ticket generation to enhance the overall user experience.

1.2 Aim of the Project

The main aim of the Smart Campus Event Management System is to develop an efficient and user-friendly platform that automates the process of organizing and

managing campus events. The project focuses on reducing manual effort, improving accessibility, and providing a seamless experience for both students and administrators. It also aims to integrate modern web technologies to deliver a secure and visually appealing application.

1.3 Scope of the Project

The scope of this project includes the development of a web-based platform for managing campus events within an educational institution. It covers functionalities such as event listing, user registration, login authentication, event participation, and ticket generation. The system is designed for students and administrators, where students can interact with event data and administrators can manage and analyze event-related activities. Future enhancements such as payment integration and mobile application support can further expand the scope of the system.

1.4 Framework

The Smart Campus Event Management System is built using a layered architecture with modern frameworks and tools.

- **Backend Framework:** Spring Boot for application development
- **Web Framework:** Spring MVC for handling requests and responses
- **Frontend Framework:** Thymeleaf for dynamic page rendering
- **Database Framework:** Spring Data JPA for database interaction
- **Security Framework:** Spring Security for authentication and authorization

The system follows the **MVC (Model-View-Controller)** design pattern, which separates business logic, user interface, and data handling for better maintainability and scalability.

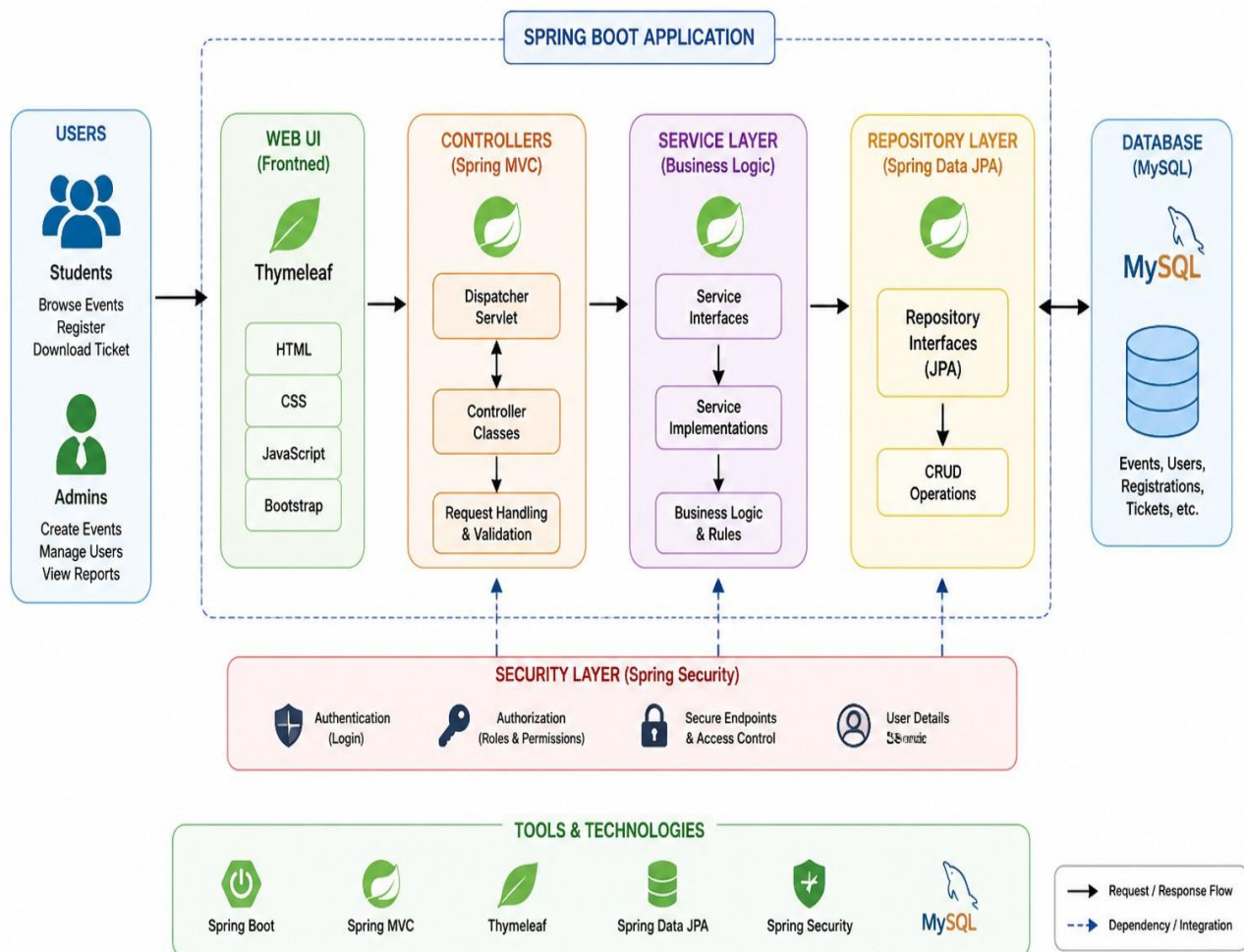


Figure 1.1: Application Framework Diagram

The above Figure 1.1 represents a layered architecture of a Spring Boot web application, showing how different components work together from the user interface to the database. Users, including students and admins, interact with the system through a web UI built with Thymeleaf, HTML, CSS, JavaScript, and Bootstrap. Their actions generate requests that are handled by the controller layer in Spring MVC, where the Dispatcher Servlet routes requests to appropriate controller classes for processing and validation. These controllers then delegate tasks to the service layer, which

contains the core business logic, including service interfaces and implementations that enforce application rules. The service layer communicates with the repository layer, implemented using Spring Data JPA, which manages data access and performs CRUD operations on the database. The database, in this case MySQL, stores all essential information such as users, events, registrations, and tickets. A security layer powered by Spring Security spans across the application, ensuring authentication, authorization, and secure access to endpoints. Overall, the diagram illustrates a clear flow of data from the user to the database and back, emphasizing separation of concerns, which enhances the application's maintainability, scalability, and organization.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

2.1 Introduction

The rapid growth of digital technologies has led to the development of various event management applications that simplify the organization and participation of events. Many existing systems provide platforms for managing registrations, scheduling events, and tracking participants. This chapter reviews some of the popular event management applications available in the market and analyzes their features, advantages, and limitations.

2.2 Existing Applications

2.2.1 Eventbrite

Eventbrite is a widely used online event management platform that allows users to create, promote, and manage events. It supports ticket sales, online registrations, and event promotion.

- Event creation and promotion

- Online ticket booking
- Payment integration
- Attendee tracking
- Complex interface for beginners
- High service charges
- Limited customization for academic use

2.2.2 Cvent

Cvent is an enterprise-level event management solution used for conferences, corporate meetings, and large-scale events.

- Advanced event planning tools
- Attendee management
- Data analytics and reporting
- Venue selection tools
- Expensive for small institutions
- Requires training to use
- Overly complex for basic campus needs

2.2.3 Meetup

Meetup is a platform designed to bring people together based on shared interests and activities.

- Group-based event creation

- Community engagement
- Event notifications
- Social interaction
- Limited administrative control
- Not suitable for formal academic events
- Less focus on structured event management

2.2.4 Whova

Whova is an event management application that focuses on enhancing attendee experience through mobile-based interaction.

- Event agenda management
- Networking tools
- Live polls and feedback
- Mobile app support
- Paid features are expensive
- Requires continuous internet connectivity
- Not tailored for campus-level usage

2.3 Comparative Analysis

Application	Strengths	Limitations
Eventbrite	Easy ticketing, popular platform	High cost, less customization
Cvent	Advanced features, analytics	Expensive, complex
Meetup	Community-based interaction	Limited admin control
Whova	Interactive features, mobile support	Costly, internet dependent

2.4 Need for Proposed System

Although several event management applications exist, they are not specifically designed for academic institutions. Most platforms are either too complex, expensive, or lack customization required for campus events. Therefore, there is a need for a dedicated system that is simple, cost-effective, and tailored to the needs of students and administrators.

The proposed Smart Campus Event Management System addresses these limitations by providing a user-friendly interface, secure login system, automated ticket generation, and real-time analytics. It is designed specifically for campus environments, making it more efficient and practical for educational institutions.

2.5 Conclusion

This chapter discussed various existing event management applications and highlighted their strengths and limitations. Based on the analysis, it is evident that there is a need for a customized solution tailored to campus requirements. The proposed system aims to overcome these challenges by providing a simplified, efficient, and cost-effective event management platform.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Smart Campus Event Management System is developed using HTML5, CSS3, and Thymeleaf template engine. The development environment includes Eclipse IDE and a web browser for testing the user interface. CSS is used to design a premium vintage gradient user interface, while Thymeleaf is used for dynamic content rendering.

3.1.2 Structure of the Project

The frontend structure is organized using templates and static resources.

- **templates/** - Contains Thymeleaf HTML pages (login, register, dashboard, events)
- **static/css/** - Stylesheets for UI design
- **static/js/** - JavaScript files for interactivity
- **static/images/** - Images used in the application

This structure ensures maintainability and easy modification of UI components.

3.1.3 Execution Procedure

The frontend is executed through the Spring Boot server. When the application runs, Thymeleaf templates are rendered dynamically based on user requests. The browser communicates with the backend via HTTP requests, and responses are displayed on the UI.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Spring Boot framework with Java. Eclipse IDE is used for development, and Maven is used for dependency management. MySQL database is used to store application data. Spring Security is integrated for authentication and authorization.

3.2.2 Structure of the Project

The backend follows a layered architecture:

- **controller/** - Handles HTTP requests and responses
- **service/** - Contains business logic
- **repository/** - Handles database operations using JPA
- **entity/** - Represents database tables
- **config/** - Security and configuration classes

This separation improves scalability and maintainability.

3.2.3 Execution Procedure

The backend starts with the Spring Boot application. When a request is received, it is processed by the controller, which interacts with the service layer. The service layer communicates with the repository to perform database operations. The processed data is then returned to the frontend. The system also includes features such as user authentication, event management, registration handling, PDF ticket generation, and admin dashboard analytics.

3.3 Database Implementation

3.3.1 Database Configuration

The Smart Campus Event Management System uses MySQL as the backend database for storing and managing application data. The database is configured using Spring Boot properties, which establish a connection between the application and the MySQL server. The configuration includes database URL, username, password, and driver class. Spring Data JPA is used to simplify database operations, and Hibernate is used as the Object Relational Mapping (ORM) tool. The configuration is defined in the `application.properties` file, enabling automatic connection and data handling.

3.3.2 Schema Structure

The database schema is designed to efficiently manage event-related data. It consists of multiple tables representing users, events, registrations, and tickets. Each table is connected through relationships such as one-to-many and many-to-one. The schema ensures data consistency and integrity by using primary keys, foreign keys, and constraints. It supports operations such as inserting new registrations, retrieving

event details, and generating reports for the admin dashboard.

3.3.3 Tables Created

The following tables are created in the database:

- **Users Table:** Stores user details such as ID, name, email, password, and role.
- **Events Table:** Contains event information such as event ID, title, description, date, location, and category.
- **Registrations Table:** Stores details of users registered for events including user ID, event ID, and registration status.
- **Tickets Table:** Contains ticket-related information such as ticket ID, event ID, user ID, and PDF file reference.

These tables work together to support the complete functionality of the system, including user authentication, event management, registration handling, and ticket generation.

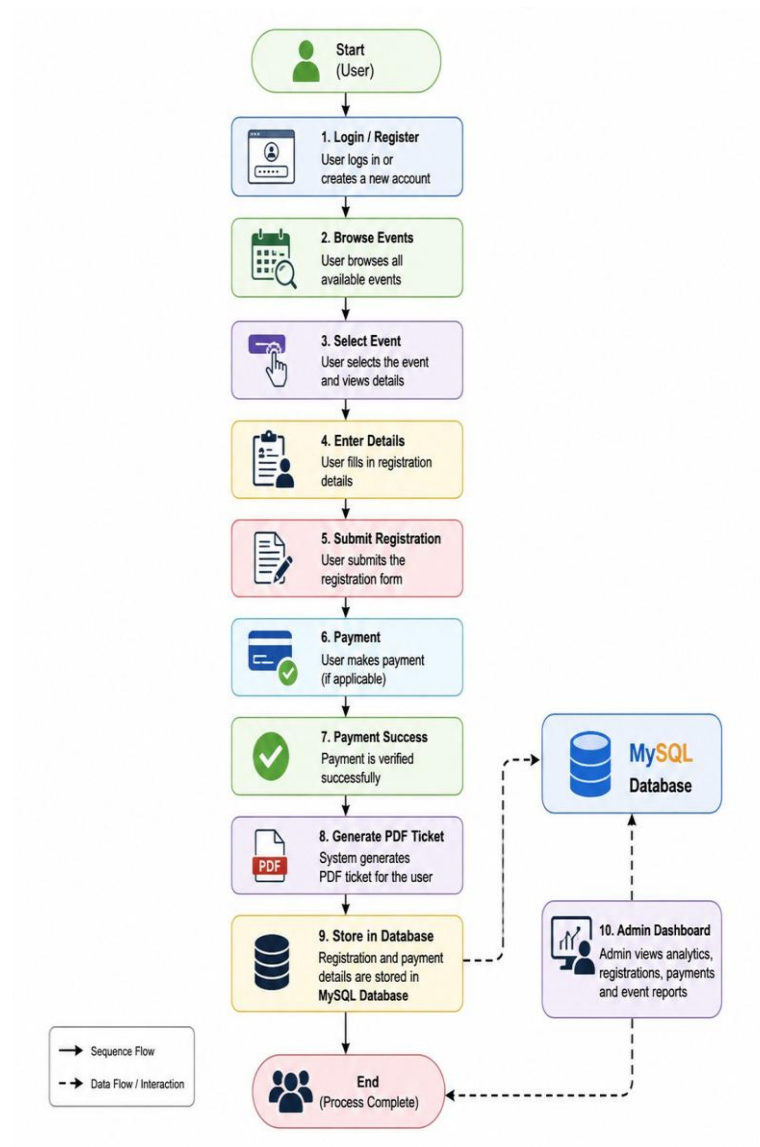


Figure 3.1: System Workflow Diagram

the above Figure 3.1 shows the system workflow where it shows all the process execution from the frontend and backend of the project and all the elements used in it.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Smart Campus Event Management System follows a layered architecture consisting of presentation, application, and database layers. The frontend is developed using HTML, CSS, and Thymeleaf, while the backend is implemented using Spring Boot. MySQL is used as the database to store user, event, and registration details.

4.2 Data Flow Diagram

The Data Flow Diagram (DFD) represents how data moves through the system. Users interact with the system by logging in, browsing events, and registering for events. The data is processed through controllers and stored in the database. The admin accesses this data for monitoring and analysis.

4.3 Module 1: User Authentication

This module handles user registration and login functionality. It ensures secure access using authentication mechanisms provided by Spring Security. Users can

create accounts, log in, and access system features.

4.4 Module 2: Event Management

This module allows administrators to create, update, and delete events. Users can browse events and view details such as date, location, and description. It ensures efficient management of event data.

4.5 Module 3: Registration and Ticket Generation

This module enables users to register for events by providing required details. After successful registration, a payment success page is displayed, and a PDF ticket is generated containing event and user details.

Advantages of this Project

- a) **User-Friendly Interface:** The system provides a clean and attractive interface for easy navigation.
- b) **Automation:** Reduces manual effort in event management and registration.
- c) **Security:** Ensures secure login and data handling using Spring Security.

Disadvantages

The system currently includes a simulated payment interface and depends on internet connectivity. Advanced features such as real-time notifications and mobile application support are not yet implemented.

Chapter 5

RESULTS AND DISCUSSIONS

The Smart Campus Event Management System was successfully developed and tested. The system provides an interactive and user-friendly interface for both students and administrators. Users can register and log into the system securely using authentication features. After logging in, users can browse available events, view detailed information, and register by entering the required personal details. Upon successful registration, a payment success page is displayed, and a PDF ticket is generated automatically. The system ensures smooth data handling and efficient performance using a MySQL database. The admin dashboard provides graphical representation of event participation using charts, helping administrators analyze user activity and event statistics effectively. The following screenshots illustrate the working of the system.

Table 5.1: System Features Comparison

Feature	Description
User Login	Secure authentication system
Event Management	Add, edit, delete events
Registration	Users can register for events
PDF Ticket	Automatic ticket generation
Admin Dashboard	Charts and analytics

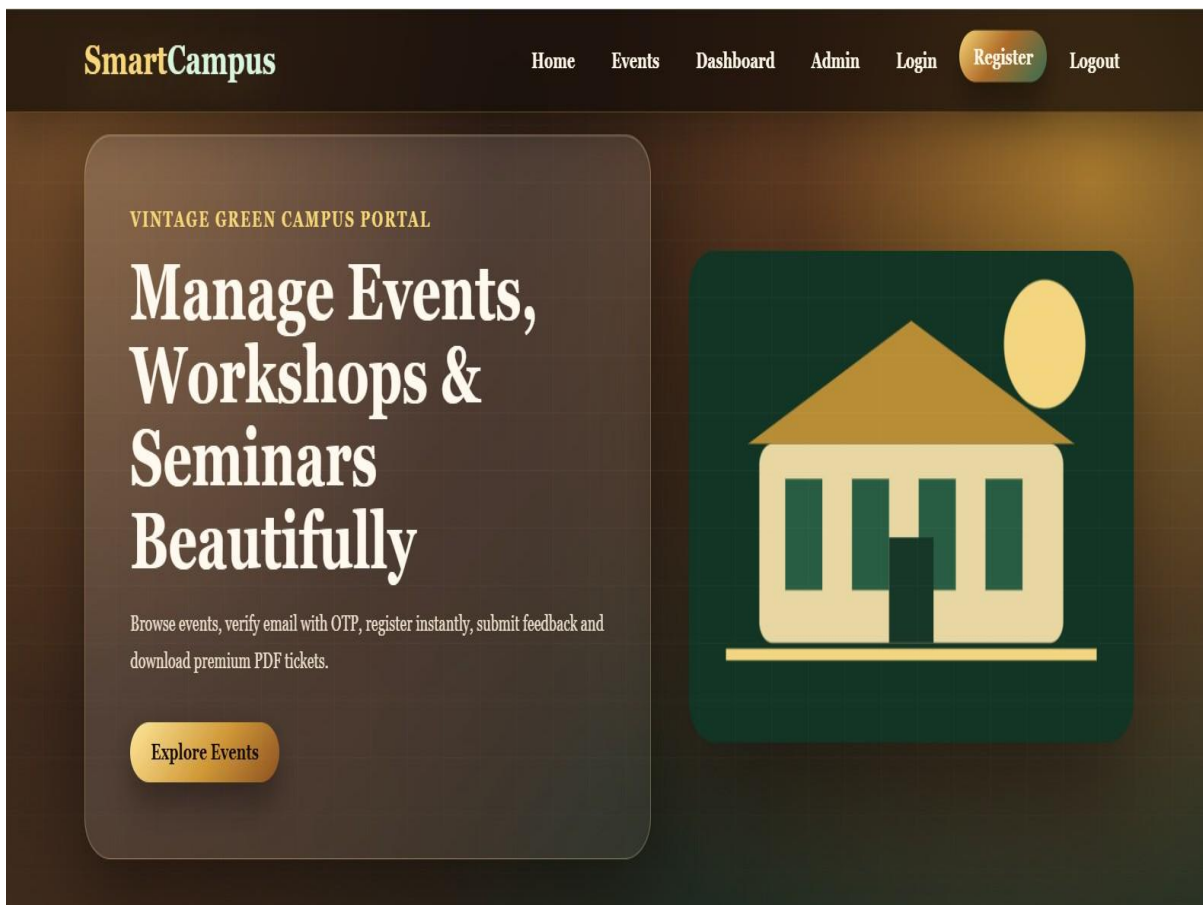


Figure 5.1: Home Page

The above Figure 5.1 shows the home page of the Smart Campus Event Management System serves as the main entry point for users. It is designed with a visually appealing and user-friendly interface that provides easy navigation to different sections of the application. The page includes a navigation bar with options such as Home, Events, Dashboard, Admin, Login, and Register. After successful login, the user's profile name and icon are displayed, enhancing personalization and usability. The central section of the homepage highlights the purpose of the system, allowing users to quickly understand the platform's functionality. A prominent "Explore Events" button is provided, which redirects users to the events listing page where they can browse available events. The design of the homepage uses a premium vintage gradient theme with smooth animations, creating an engaging user experience. The layout is responsive and accessible across different devices. Overall, the homepage acts as a gateway to the system, enabling users to navigate efficiently and access event-related features seamlessly.

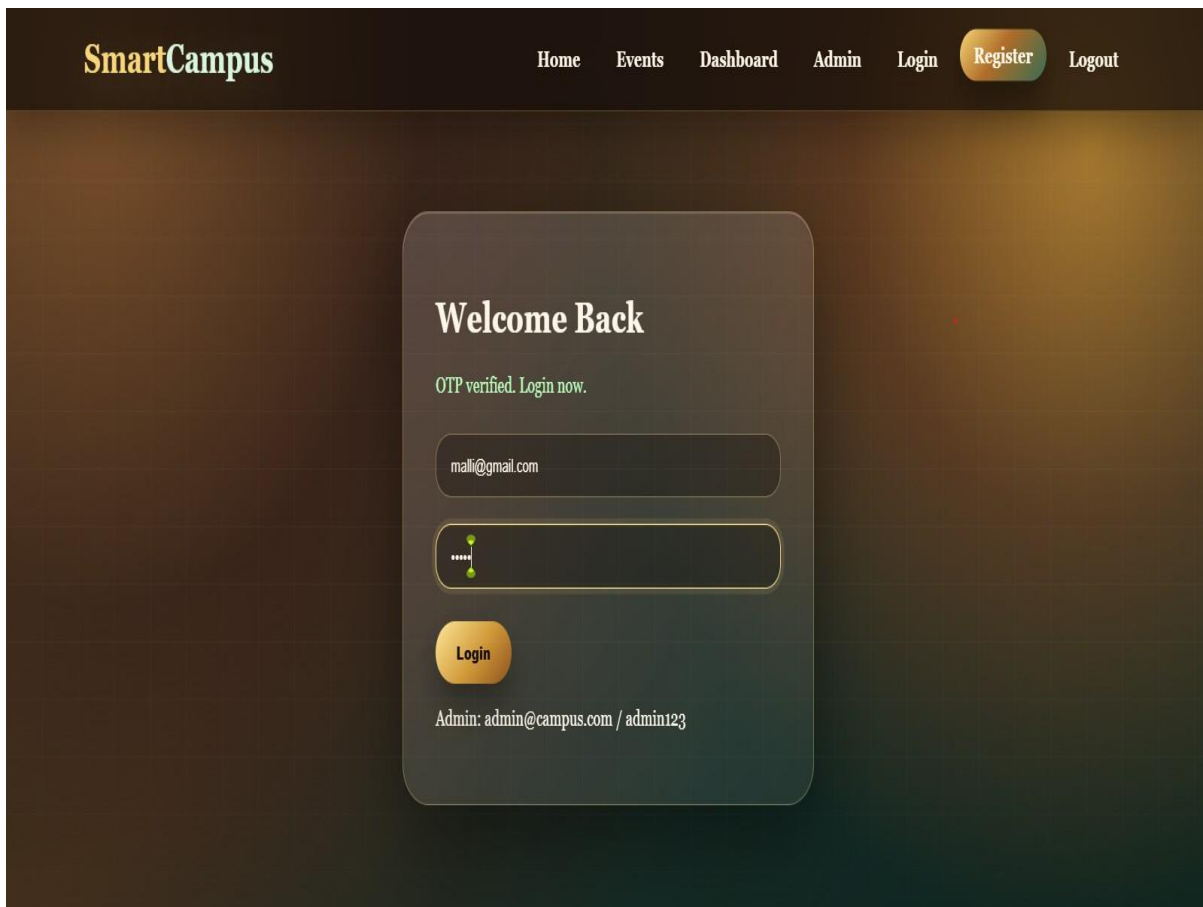


Figure 5.2: Login Page

The above figure 5.2 shows the login page of the Smart Campus Event Management System provides a secure interface for users to access their accounts. It allows registered users to enter their credentials, such as email and password, to authenticate and log into the system. The login form includes proper validation to ensure that only valid inputs are accepted. If incorrect credentials are entered, the system displays an error message, preventing unauthorized access. The authentication process is handled using Spring Security, ensuring data protection and secure session management. The page is designed with a clean and visually appealing layout that matches the overall vintage theme of the application. It also includes navigation options such as “Register” for new users and “Home” for easy redirection. Once the login is successful, users are redirected to the dashboard or events page, where they can explore available events and perform further actions.

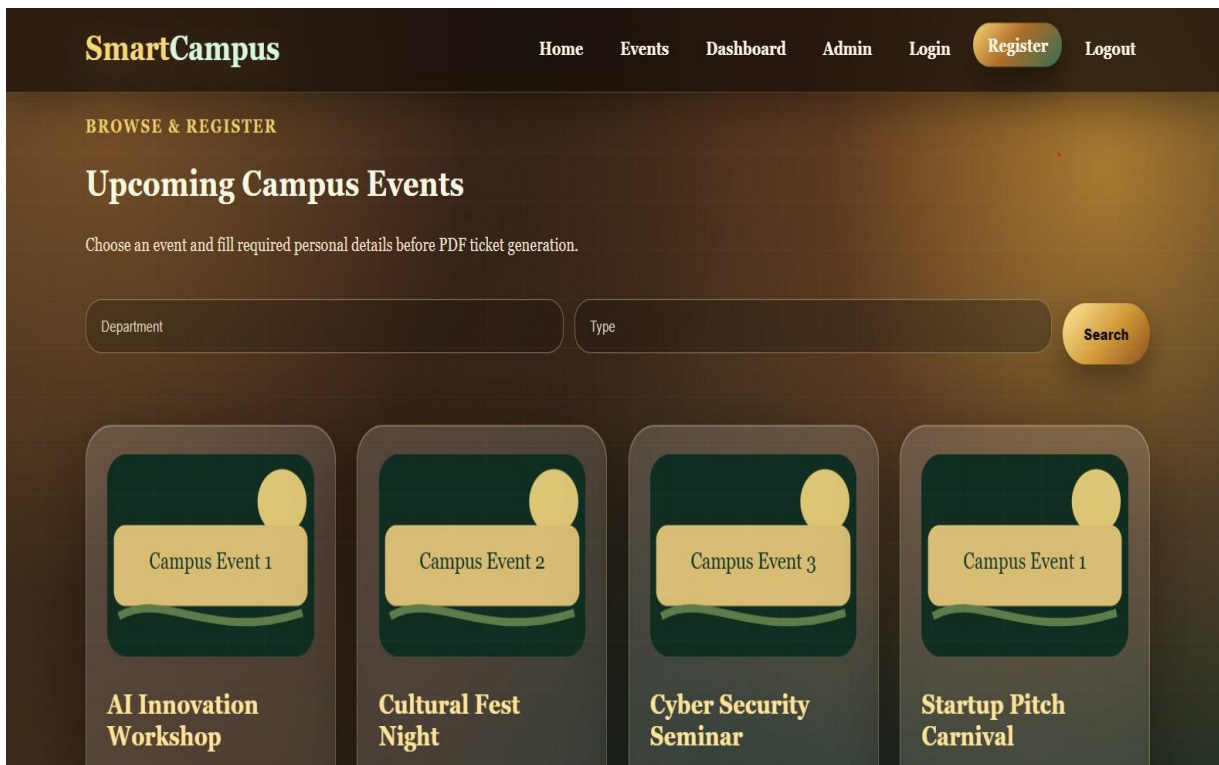


Figure 5.3: Events Page

The above Figure 5.3 The events page of the Smart Campus Event Management System displays a list of all available campus events in a structured and user-friendly format. It allows users to browse through different events such as workshops, seminars, and technical sessions. Each event is presented with essential details including the event title, date, location, and a brief description. The page is dynamically generated using Thymeleaf, ensuring that all event data is fetched from the MySQL database and updated in real time. Users can easily select an event of interest and view more details or proceed to register for that event. A dedicated “Register” or “View Details” button is provided for each event, enabling smooth navigation to the registration process. The design of the events page follows the premium vintage theme of the application, with well-aligned cards, smooth animations, and responsive layout. This enhances readability and improves the overall user experience. Overall, the events page acts as a central hub where users can explore and interact with all available events efficiently.

The screenshot shows the 'SmartCampus' website with a dark theme. The navigation bar includes links for Home, Events, Dashboard, Admin, Login, Register, and Logout. The main content area is titled 'PREMIUM EVENT REGISTRATION' and features a section for 'Cultural Fest Night'. Below the title, a message states: 'Fill all required details. These details will be printed on your downloadable PDF ticket.' The registration form consists of several input fields: 'Name' (containing 'mali'), 'Email' (containing 'mali@gmail.com'), 'Phone Number *', 'Roll Number / ID *', 'Department *', 'Academic Year / Semester', and a numeric field for the number of participants (containing '1'). There is also a text area for 'Special requirements / notes'. At the bottom of the form, there are two buttons: 'Confirm Registration' and a link 'Back to Events'.

Figure 5.4: Events Page

The student event registration page allows users to register for selected events by providing their personal and event-related details. This page is accessed after the user selects an event from the events listing page. The form typically includes fields such as name, email, phone number, number of participants, and any additional information required for the event. Proper input validation is implemented to ensure that all mandatory fields are filled correctly before submission. Once the user submits the form, the system processes the registration details and stores them securely in the MySQL database. A confirmation or payment success interface is then displayed to indicate successful registration. The page is designed with a clean and structured layout, following the overall premium vintage theme of the application. It ensures ease of use with clearly labeled fields, responsive design, and smooth navigation. Overall, the student event registration page plays a crucial role in capturing user information and enabling seamless participation in campus events.

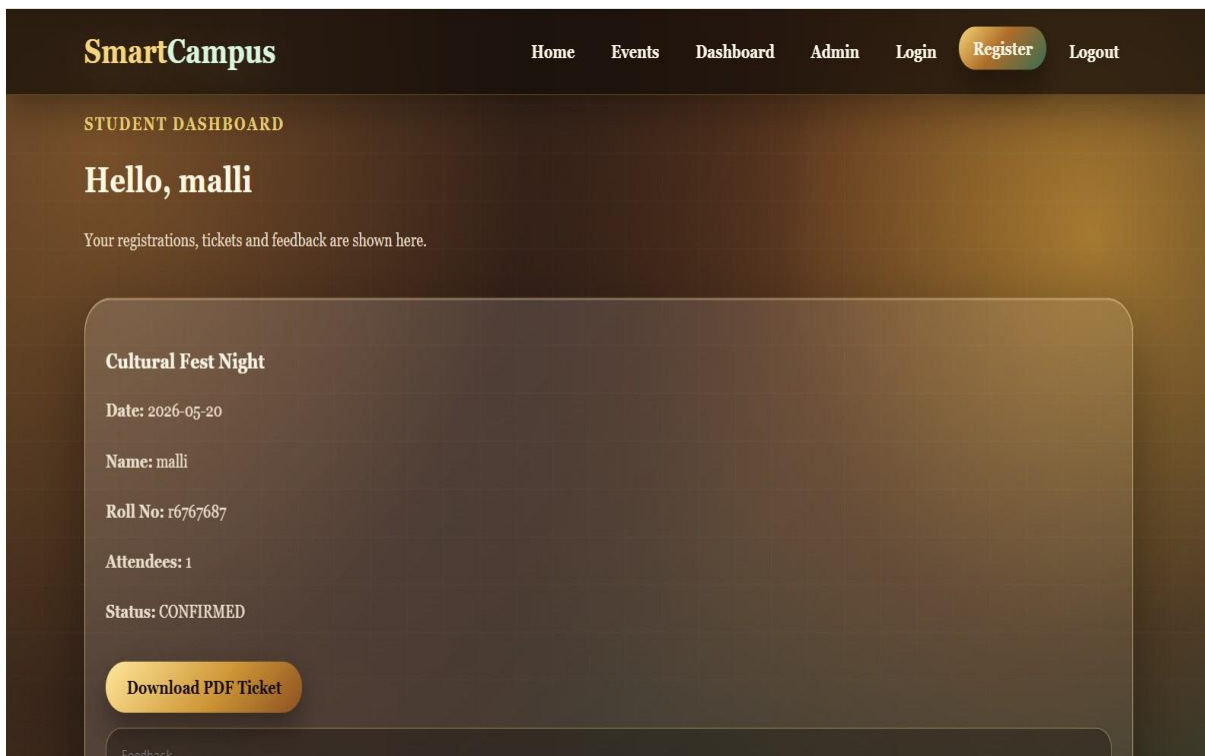


Figure 5.5: Booking Status Confirmation

The booking status feature in the Smart Campus Event Management System allows users to track the current state of their event registrations. After completing the registration process, each booking is assigned a status such as Pending, Confirmed, or Completed. When a user submits the registration form, the status is initially set to Pending. Once the registration is successfully processed and verified (including payment confirmation if applicable), the status is updated to Confirmed. This helps users clearly understand whether their registration has been accepted. The booking status is stored in the MySQL database and is dynamically retrieved whenever the user views their registration details or dashboard. This ensures that users always have access to the latest information regarding their bookings. The feature enhances transparency and improves user experience by providing real-time updates on registration progress. It also assists administrators in managing event participation efficiently by monitoring and updating booking statuses when required.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Smart Campus Event Management System was successfully designed and implemented to simplify the process of managing campus events. The system provides an efficient platform for students to browse events, register online, and download tickets, while administrators can manage events and monitor participation effectively. The application integrates modern technologies such as Spring Boot, Thymeleaf, and MySQL to ensure scalability, security, and performance. Features such as user authentication, event registration, PDF ticket generation, and admin dashboard analytics enhance the overall functionality of the system. The developed system reduces manual workload, improves accessibility, and provides a user-friendly interface. It demonstrates how web-based technologies can be effectively used to automate real-world processes in an academic environment.

6.2 Future Enhancements

The system can be further improved by incorporating advanced features and technologies. Some of the possible future enhancements are listed below:

- Integration of real-time payment gateway for secure online transactions
- Implementation of email and SMS notifications for event updates
- Addition of QR code verification for event entry validation
- Development of a mobile application for better accessibility
- AI-based event recommendation system for users
- Real-time chat or communication feature between users and admins

Review of Existing Applications

Existing event management platforms such as Eventbrite, Meetup, and Cvent provide advanced features for event organization and management. However, these applications are often complex, costly, and not specifically designed for campus-level usage.

Eventbrite (<https://www.eventbrite.com>) provides ticketing and event promotion features but involves service charges. Meetup (<https://www.meetup.com>) focuses on community-based events but lacks structured administrative control. Cvent (<https://www.cvent.com>) offers enterprise-level solutions but is expensive and complex for small institutions.

Compared to these platforms, the proposed Smart Campus Event Management System provides a simple, cost-effective, and customized solution specifically designed for educational institutions.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of knowledge	Requires in depth engineering knowledge	WK3, WK4, WK5, WK6	Requires knowledge in Spring Boot, MVC architecture, database design, REST APIs, and frontend technologies for developing a scalable event management system.
WP2	Conflicting requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances usability and security by ensuring easy student access while maintaining secure admin authentication and data validation.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	Involves system design, database schema creation, validation handling, and efficient event filtering and registration mechanisms.
WP4	Familiarity	Partially new problems	WK4, WK8	Integrating real-time event updates, user interaction, and admin analytics introduces challenges beyond traditional manual systems.
WP5	Applicable Codes	Limited standard solutions	WK6	Requires custom implementation for event filtering, registration tracking, and statistics generation not fully covered by standard templates.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Addresses requirements of students, faculty coordinators, and administrators, ensuring smooth event organization and participation.
WP7	Interdependence	System level integration	WK3, WK5, WK6 25	Integrates frontend UI, backend services, database, validation, and security modules into a unified system.

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes Spring Boot, Spring MVC, Spring Data JPA, Thymeleaf, HTML/CSS, and relational databases for full-stack development.
EA2	Level of Interactions	Complex interactions between modules	Manages interactions between event management, user registration, validation, authentication, and reporting modules.
EA3	Innovation	Application of modern technologies	Implements a web-based centralized platform with filtering, real-time updates, and secure admin operations.
EA4	Consequences to Society and Environment	Impact on users and society	Enhances student engagement and improves institutional efficiency while reducing manual errors and delays.
EA5	Familiarity	Beyond standard practices	Combines web development, security, and data analytics into a unified system, extending beyond traditional event handling methods.

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Facilitates student participation in academic events, workshops, and seminars, enhancing learning beyond the classroom.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in educational institutions through a centralized event management system.
SDG 11	Sustainable Cities and Communities	Supports smart campus initiatives by digitizing event processes and improving resource management.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent event handling through authentication and controlled access.
SDG 12	Responsible Consumption and Production	Reduces paper usage by enabling online event registration and management.

Table 7.3: SDG Mapping for the Project

SOURCE CODE

```
package com.smartcampus.controller;

import com.smartcampus.model.Event;
import com.smartcampus.service.EventService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/events")
public class EventController {

    @Autowired
    private EventService eventService;

    // Get all events
    @GetMapping
    public List<Event> getAllEvents() {
        return eventService.getAllEvents();
    }

    // Add new event
    @PostMapping
    public Event createEvent(@RequestBody Event event) {
        return eventService.saveEvent(event);
    }

    // Delete event by ID
    @DeleteMapping("/{id}")
    public String deleteEvent(@PathVariable Long id) {
        eventService.deleteEvent(id);
        return "Event deleted successfully!";
    }
}
```

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